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SmartHire — Campus Recruitment System

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ABSTRACT

- **Background:** Campus placement cycles represent high-concurrency recruitment operations engaging students, company recruiters, and institutional Training & Placement Officers (TPOs).
- **The Problem:** Traditional workflows rely on scattered Google Forms, manual spreadsheet trackers, and unproctored assessment links, creating severe communication bottlenecks and fraud risks.
- **SmartHire Solution:** A unified cloud recruitment ecosystem pairing role-based placement tracking with client-side proctoring, multi-language coding sandboxes, and transparent career readiness modules.

INTRODUCTION

- **Tripartite Stakeholders:** Unifies *Students* (drive tracking & skill practice), *Recruiters* (requisitions & candidate reviews), and *TPOs* (compliance auditing).
- **Proctored Assessments:** Ingests 32 corporate mock OA suites (86 verified problems) with client-side anti-cheat telemetry (tab switch, face presence, clipboard locking).
- **Ethical Human-in-the-Loop Gate:** Unlike black-box AI platforms that auto-reject candidates, SmartHire isolates ATS heuristic scoring strictly as formative guidance for students.

LITERATURE SURVEY

Dimension	Legacy Systems	Commercial ATS	SmartHire (Ours)
Architecture	Monolithic PHP / Relational DBs	Proprietary Closed SaaS	**Decoupled React 19 + Express + Cloud DB**
Online Coding OA	None (Third-party links)	Paywalled External	**Integrated Sandboxed Multi-Lang (JDoodle)**
Exam Proctoring	None (Unsupervised)	Bandwidth-heavy Video Streams	**Privacy-Preserving Client Telemetry (<12 kbps)**
ATS Resume Engine	None / Manual Sorting	Opaque AI Auto-Rejection	**100-Pt Transparent Heuristic (Advisory)**
Drive Alerts	Email / Notice Boards	Email Digest (Delayed)	**Realtime Handheld Push Alerts (<315 ms)**
Candidate Hiring	Manual Spreadsheets	Automated Black-Box Filters	**Enforced Human-in-the-Loop Selection**

Table 1: Architectural comparison with published recruitment and assessment literature.

METHODOLOGY / IMPLEMENTATION

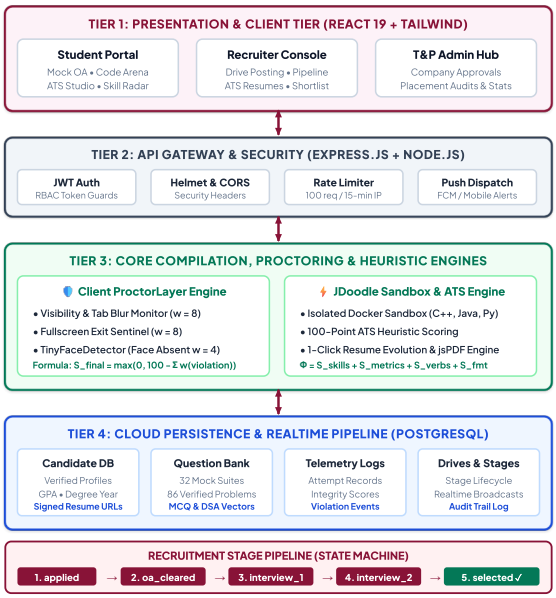


Fig 1: SmartHire Multi-Tier System Architecture & Stage Machine Workflow

100-Point Calibrated ATS Formulation

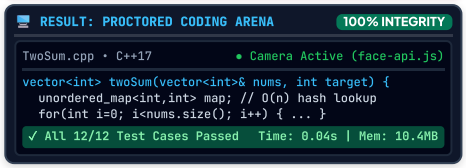
$\Phi = S_{skills}(35) + S_{metrics}(15) + S_{verbs}(15) + S_{arch}(15) + S_{vocab}(10) + S_{format}(10)$

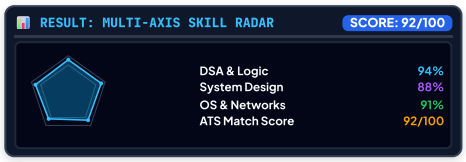
Calculated entirely in-browser using symbol-aware matching for non-alphanumeric tokens (e.g., `C++`, `Node.js`).

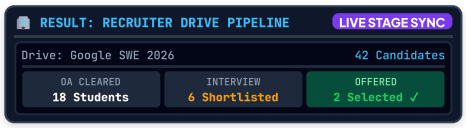
Privacy-Preserving Practice Proctoring Telemetry

$S_{final} = \max(0,\ 100 - \Sigma\ w(violation))$: w(tab_switch) = 8, w(fullscreen_exit) = 8, w(multiple_faces) = 12, w(paste) = 6. Replaces video uplinks with <12 kbps discrete event payloads.

RESULTS & ANALYSIS







172ms	>94%	315ms	100%
CLIENT ATS LATENCY	BANDWIDTH SAVED	PUSH ALERT SYNC	RBAC SECURITY

CONCLUSION & FUTURE SCOPE

- **Conclusion:** SmartHire replaces disparate placement tools with a secure, automated recruitment platform ensuring complete meritocracy and privacy.
- **Future Scope 1:** AI-powered Voice & NLP Mock Interviewer evaluating conversational clarity, body language, and system design articulation.
- **Future Scope 2:** Cross-platform native mobile application with offline practice suites and automated alumni mentorship connections.

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